11.1.6 Wrap-Up: Reflection

1. Complete the table.

The terms and information used in the lesson that I knew were …	Something I wondered during the lesson was …	Something I learned during the lesson was …

Unit 11, Lesson 2: Key Features of Exponential Functions

Benchmarks

MA.912.AR.5.3 Given a mathematical or real-world context, classify exponential functions as representing growth or decay.

MA.912.AR.5.6 Given a table, equation, or written description of an exponential function, graph that function and determine its key features.

Learning Targets

- ☐ I can determine the key features of an exponential function from a table, graph, or exponential function.
- ☐ I can classify a function as representing exponential growth or decay.

11.2.1 Warm-Up: Rhind Papyrus

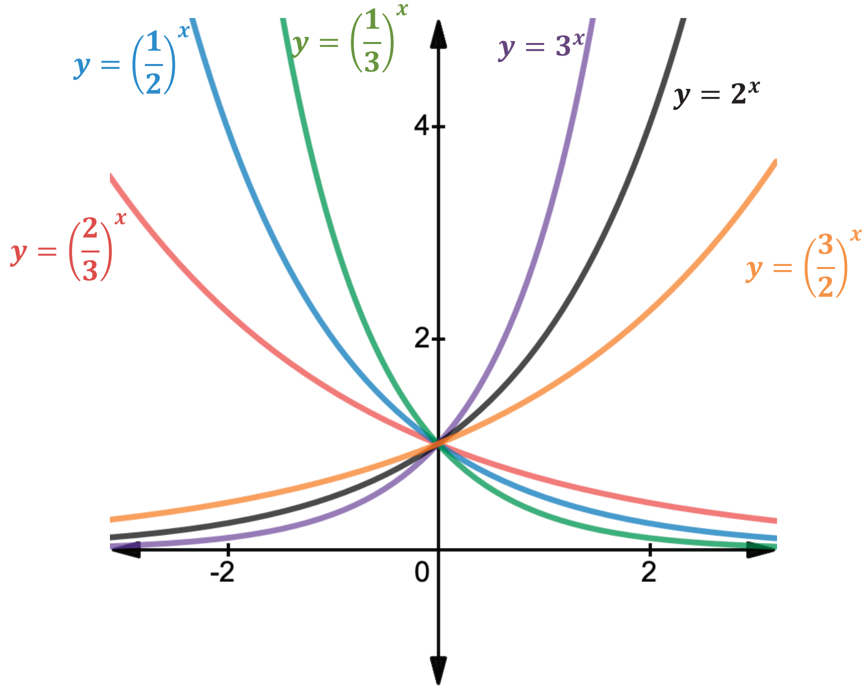
Some of the earliest records of mathematical problem solving come from ancient Egypt in the form of papyri that were written between 1850 and 1600 BCE. One of these, the Rhind Mathematical Papyrus, is a collection of math problems written around 1650 BCE. The problems in the papyrus were used to manage food supplies for cities, among other things.

One of the problems asks: "In a certain village, there were 7 houses; each house had 7 cats; each cat caught 7 mice; each mouse would have (were it not for the cats) eaten 7 ears of spelt; each ear of spelt produced 7 hekats of grain at harvest. How many hekats of grain were saved by the presence of the cats?"

1. Find the solution to the Rhind Papyrus problem.

11.2.2 Exploration: Equations, Graphs, and Tables of Exponential Functions

1. The graphs of six exponential functions are shown.



a. Work with your partner to complete the table.

	Graphs with a *b*-value Between 0 and 1	**Graphs with a *b*-value Greater Than 1**
Equations	☐ $y = \left(\frac{2}{3}\right)^x$ ☐ $y = \left(\frac{1}{2}\right)^x$ ☐ $y = \left(\frac{1}{3}\right)^x$ ☐ $y = 3^x$ ☐ $y = 2^x$ ☐ $y = \left(\frac{3}{2}\right)^x$	☐ $y = \left(\frac{2}{3}\right)^x$ ☐ $y = \left(\frac{1}{2}\right)^x$ ☐ $y = \left(\frac{1}{3}\right)^x$ ☐ $y = 3^x$ ☐ $y = 2^x$ ☐ $y = \left(\frac{3}{2}\right)^x$
Type of Function	○ Exponential growth ○ Exponential decay	○ Exponential growth ○ Exponential decay

b. Complete the statements.

For the equations that represent exponential growth, as the x-values increase, the y-values
- ○ increase.
- ○ approach zero.

For the equations that represent exponential decay, as the x-values increase the y-values
- ○ increase.
- ○ approach zero.

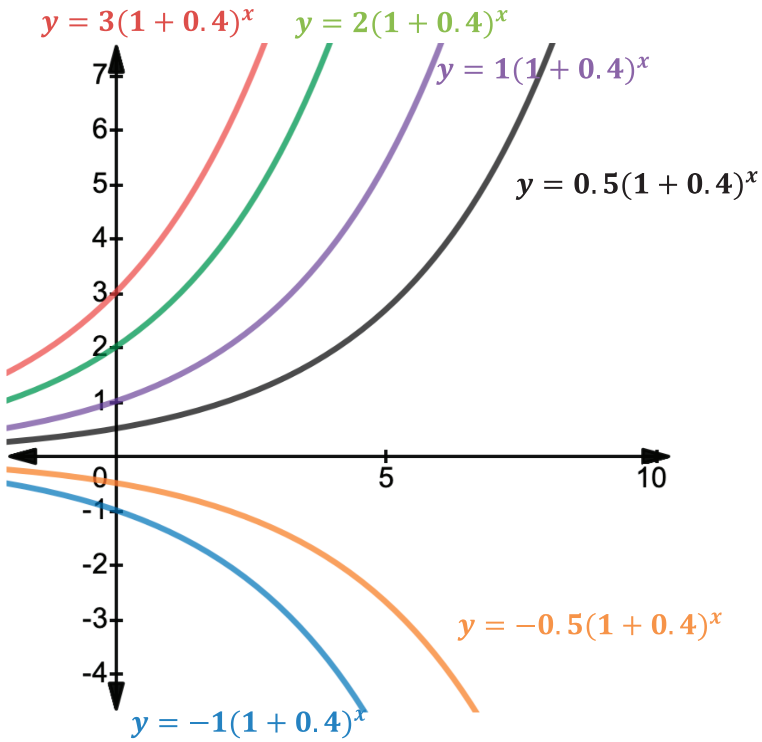
An ***asymptote*** is a line that a graph approaches, but never crosses or touches.

c. For the exponential decay equations, as x approaches $+\infty$, what is the horizontal asymptote?

d. For the exponential growth equations, as x approaches $-\infty$, what is the horizontal asymptote?

e. What do you notice about the graphs of exponential decay and exponential growth functions as the value of b gets closer to 1?

2. The graphs of six exponential functions are shown with their corresponding equations.



a. What do all the equations have in common?

b. What do you notice as the value of a approaches 0?

c. Complete the table to describe the similarities and differences of the graphs of $y = -1(1 + 0.4)^x$ and $y = 1(1 + 0.4)^x$.

Similarities	Differences

11.2.3 Bring it Together: Key Features of Exponential Functions

1. Based on the 12 graphs in the Exploration, complete the table.

Key Feature	Question
Domain	Do all of the graphs have a domain of all real numbers? ○ yes ○ no
Range	Do all of the graphs have a range of all real numbers? ○ yes ○ no
x-intercepts	Do the graphs show an x-intercept? ○ yes ○ no
y-intercept	Do the graphs show a y-intercept? ○ yes ○ no
Vertex	Do the graphs show a vertex? ○ yes ○ no
Axis of Symmetry	Do the graphs show an axis of symmetry? ○ yes ○ no

Increasing/ Decreasing	
	Circle the graphs that show intervals where the function is increasing.

$y = \left(\frac{2}{3}\right)^x$	$y = \left(\frac{1}{2}\right)^x$	$y = \left(\frac{1}{3}\right)^x$
$y = (3)^x$	$y = (2)^x$	$y = \left(\frac{3}{2}\right)^x$
$y = 3(1 + 0.4)^x$	$y = 2(1 + 0.4)^x$	$y = 1(1 + 0.4)^x$
$y = 0.5(1 + 0.4)^x$	$y = -0.5(1 + 0.4)^x$	$y = -1(1 + 0.4)^x$

Circle the graphs that show intervals where the function is decreasing.

$y = \left(\frac{2}{3}\right)^x$	$y = \left(\frac{1}{2}\right)^x$	$y = \left(\frac{1}{3}\right)^x$
$y = (3)^x$	$y = (2)^x$	$y = \left(\frac{3}{2}\right)^x$
$y = 3(1 + 0.4)^x$	$y = 2(1 + 0.4)^x$	$y = 1(1 + 0.4)^x$
$y = 0.5(1 + 0.4)^x$	$y = -0.5(1 + 0.4)^x$	$y = -1(1 + 0.4)^x$

Positive/ Negative	
	Circle the graphs that show intervals where the function is positive.

$y = \left(\frac{2}{3}\right)^x$	$y = \left(\frac{1}{2}\right)^x$	$y = \left(\frac{1}{3}\right)^x$
$y = (3)^x$	$y = (2)^x$	$y = \left(\frac{3}{2}\right)^x$
$y = 3(1 + 0.4)^x$	$y = 2(1 + 0.4)^x$	$y = 1(1 + 0.4)^x$
$y = 0.5(1 + 0.4)^x$	$y = -0.5(1 + 0.4)^x$	$y = -1(1 + 0.4)^x$

Circle the graphs that show intervals where the function is negative.

$y = \left(\frac{2}{3}\right)^x$	$y = \left(\frac{1}{2}\right)^x$	$y = \left(\frac{1}{3}\right)^x$
$y = (3)^x$	$y = (2)^x$	$y = \left(\frac{3}{2}\right)^x$
$y = 3(1 + 0.4)^x$	$y = 2(1 + 0.4)^x$	$y = 1(1 + 0.4)^x$
$y = 0.5(1 + 0.4)^x$	$y = -0.5(1 + 0.4)^x$	$y = -1(1 + 0.4)^x$

End Behavior	What do you notice about the end behavior as x-values get smaller? What do you notice about the end behavior as x-values get larger?
Asymptotes	What is the horizontal asymptote of exponential growth functions, $f(x) = a(b)^x$ or $f(x) = a(1 + r)^x$? What is the horizontal asymptote of exponential decay functions, $f(x) = a(b)^x$ or $f(x) = a(1 - r)^x$?

11.2.4 Guided Instruction: Determining Key Features of Exponential Graphs

A table and graph of the exponential function $B(x) = 3(1 - 0.8)^x$ are shown. The key features of the graph are color-coded with the key features in the table.

Graph

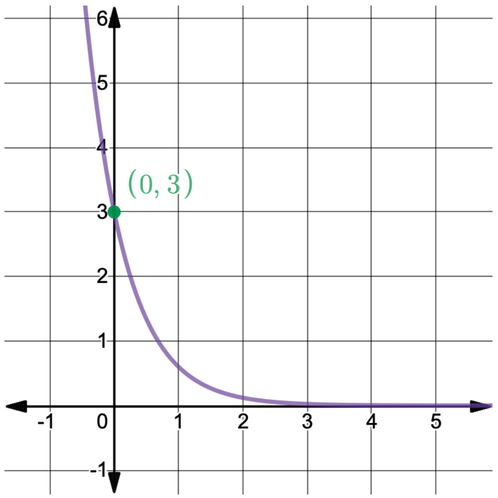


Table of Values

x	y	
-2	75	$\times\ 20\%$
-1	15	$\times\ 20\%$
0	3	$\times\ 20\%$
1	0.6	$\times\ 20\%$
2	0.12	

Key Features

Domain	$\mathbb{R}$	**Range**	$y > 0$
y-intercept	$(0,3)$	**Constant Percent Rate of Change**	$B(x) = 3(1 - 0.8)^x$ $B(x) = 3(0.2)^x$ $0.2 \cdot 100 = 20\%$
Increasing or Decreasing	Decreasing $-\infty < x < \infty$	**Positive**	$-\infty < x < \infty$
		Negative	Never negative
Left End Behavior	$x \to -\infty,\ y \to \infty$	**Right End Behavior**	$x \to \infty,\ y \to 0$

1. Describe how the constant percent rate of change is found from the table and function.

2. Use the graph and table of $y = -2\left(\frac{1}{2}\right)^x$ to determine the key features.

Graph

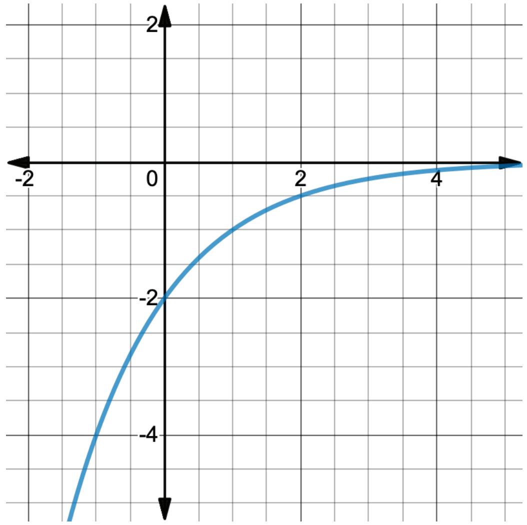


Table of Values

x	y
-2	-8
-1	-4
0	-2
1	-1
2	$-\frac{1}{2}$

Key Features

Domain		**Range**	
y-intercept		**Constant Percent Rate of Change**	
Increasing or Decreasing		**Positive**	
		Negative	
Left End Behavior		**Right End Behavior**	

3. Use the graph and table of $y = -(1 + 0.75)^x$ to determine the key features.

Graph

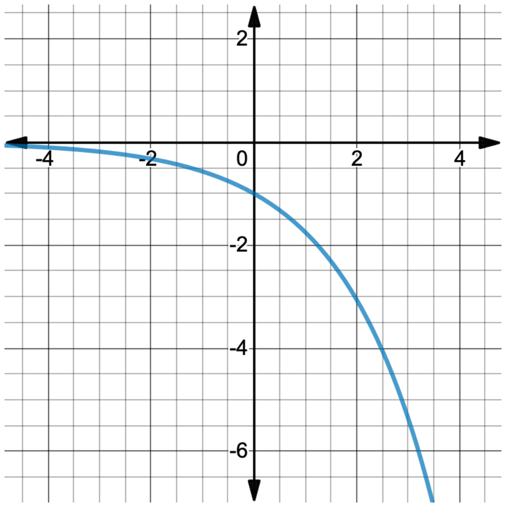


Table of Values

x	y
-2	$\frac{-16}{49}$
-1	$\frac{-4}{7}$
0	-1
1	$\frac{-7}{4}$
2	$\frac{-49}{16}$

Key Features

Domain		**Range**	
y-intercept		**Constant Percent Rate of Change**	
Increasing or Decreasing		**Positive**	
		Negative	
Left End Behavior		**Right End Behavior**	

4. Use the graph and table of $B(x) = 3\left(\frac{7}{4}\right)^x$ to determine the key features.

Graph

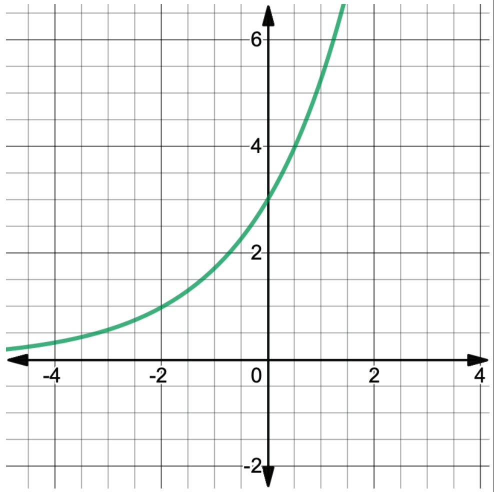


Table of Values

x	y
-2	$\frac{48}{49}$
-1	$\frac{12}{7}$
0	3
1	$\frac{21}{4}$
2	$\frac{147}{16}$

Key Features

Domain		**Range**	
y-intercept		**Constant Percent Rate of Change**	
Increasing or Decreasing		**Positive**	
		Negative	
Left End Behavior		**Right End Behavior**	

5. Using the key features found in the previous problems, complete the table.

a is positive or negative	Constant Percent Rate of Change	Increasing or Decreasing	End Behavior as x Increases	End Behavior as x Decreases
Positive	Greater than 100%			
Positive	Between 0% and 100%			
Negative	Greater than 100%			
Negative	Between 0% and 100%			

6. Discuss with your partner what effect the a-value and the constant percent rate of change have on the graph. Write a summary of your discussion.

11.2.5 Your Turn!

1. Use the graph and table of $y = -7(1 - 0.1)^x$ to determine the key features.

Graph

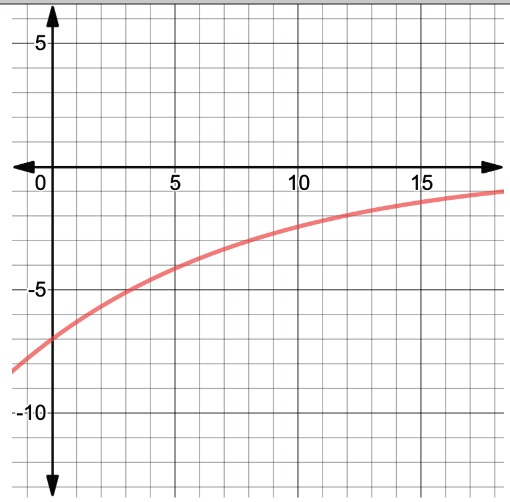


Table of Values

x	y
-2	$-\frac{700}{81}$
-1	$-\frac{70}{9}$
0	-7
1	$-\frac{63}{10}$
2	$-\frac{567}{100}$

Key Features

Domain		**Range**	
y-intercept		**Constant Percent Rate of Change**	
Increasing or Decreasing		**Positive**	
		Negative	
Left End Behavior		**Right End Behavior**	

11.2.6 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each key feature.

I need help. I got it. I could help someone.

Finding domain and range of an exponential function	
Finding the constant percent rate of change	
Describing the end behavior of an exponential function	
Finding the y-intercept	
Finding the intervals where the function is positive or negative	
Finding the intervals where the function is increasing or decreasing	
Identifying the horizontal asymptote	

Unit 11, Lesson 3: Graphing Exponential Functions Given an Equation

Benchmarks

MA.912.AR.5.3 Given a mathematical or real-world context, classify exponential functions as representing growth or decay.

MA.912.AR.5.6 Given a table, equation, or written description of an exponential function, graph that function and determine its key features.

Learning Targets

- ☐ I can graph an exponential function given the equation.
- ☐ I can classify a function as representing exponential growth or decay.